

Unlocking E-Commerce Insights

Hindi Product Review Sentiment Analyser

THE GAP

While millions shop online in India, most commercial sentiment tools are built strictly for English. This leaves D2C brands and small businesses completely blind to the actionable insights hidden inside Hindi (Devanagari) customer feedback.

THE SOLUTION

An end-to-end Machine Learning pipeline and interactive web application that reads Devanagari product reviews and categorizes them as **Positive** or **Negative**, complete with calibrated confidence scores.

THE ENGINE & PERFORMANCE

We built two approaches to balance cost, speed, and accuracy:

- **The Lightweight Champion (IndicBERT):** A 33M parameter, ALBERT-based model fine-tuned specifically on Indic datasets. Achieved a solid **82.76% test accuracy** (0.8274 Macro-F1).
- **The Heavyweight Baseline (LLaMA 3.3 70B):** Zero-shot inference via Groq's LPU API. Achieved a perfect **100% accuracy** on the test set.

THE ENGINEERING TRADE-OFF

LLaMA 3.3 is flawless but relies on external API calls, introducing latency and recurring costs, and is **2,121x larger** than our custom model.

APP FEATURES

- **Real-Time Analysis:** Single review checks with confidence breakdowns.
- **Batch Processing:** Bulk CSV uploads with downloadable reports.
- **Insights Dashboard:** Automated sentiment distribution and *Zero-Inference Cost* Top Theme detection (e.g., Quality, Delivery, Price).

Our fine-tuned **IndicBERT** model is only **122MB**, runs locally with **sub-100ms latency**, and completely eliminates recurring API costs, making it highly practical and cost-effective for production deployment.

 Check out the code: github.com/RajaiTarun/Hindi-product-review-sentiment